

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5115

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 2

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Nalluri Charan Sai (192372003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: NALLURI CHARAN SAI

1) Suspicious Digital Asset Investigation

Scenario: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

Concealing data inside a digital image falls under steganography, which is fundamentally different from encryption: encryption scrambles a message so it cannot be read, whereas steganography hides the very existence of the message so an observer does not even suspect that secret content is present. Because the logo 'appears normal upon casual inspection', a purely visual check by a human eye is exactly the situation steganography is designed to defeat — the cover image is deliberately chosen or crafted so that embedding leaves no perceptible visual trace. Forensic investigators must therefore go beneath the surface and apply a combination of comparative, statistical, structural, and tool-based techniques to confirm or rule out hidden content.

(a) How Investigators Could Detect the Presence of Hidden Data

A forensic investigation into a suspected steganographic image typically proceeds through several complementary layers of analysis, since no single test is conclusive on its own:

1. File property and metadata comparison: The investigator first compares the suspect logo's file size, cryptographic hash value (MD5/SHA-256), pixel resolution, colour depth, and creation/modification timestamps against a known-clean copy of the same logo obtained from an official archive or the design team. A steganographic payload adds extra bits to the cover file, so an unusually large file size for a simple flat-color logo, or a hash value that does not match the verified original, is one of the fastest and most reliable red flags even before any deep analysis is performed.

2. Statistical / Least Significant Bit (LSB) analysis: The most common image-steganography method hides secret bits in the least significant bit of each pixel's colour channel, since changing this bit alters the colour value by at most 1 out of 256 levels — invisible to the human eye. Investigators extract and visually enhance the LSB plane of every color channel; in an unmodified image the LSB plane looks like fine random noise that correlates weakly with the image content, but in a steno-image carrying structured hidden data (such as text or another file), the LSB plane often shows unnatural patterns, blocky regions, or repeating structure that betrays the embedding.

3. Chi-square and histogram (statistical) analysis: Because LSB embedding forces pairs of adjacent pixel values (called Pairs of Values, or POVs) to occur with near-equal frequency, a chi-square test can statistically detect this artificial balancing even when it is invisible to the eye. Similarly, plotting the colour histogram of the suspect image and comparing it with the histogram of a genuine, unmodified logo can reveal unnatural smoothing, unexpected spikes at specific intensity values, or a 'comb-like' pattern — all classic statistical fingerprints of LSB-based data hiding.

4. Signature and file-structure analysis: Using a hex editor, investigators inspect the raw byte structure of the file, paying particular attention to header fields, chunk/segment boundaries (e.g., PNG IDAT/IEND chunks or JPEG marker segments), and the region after the legitimate End-Of-File (EOF) marker. Many amateur steganography tools simply append the hidden payload after the image's real EOF marker; most image viewers ignore this trailing data and display the logo normally, so its presence is invisible to a casual viewer but is immediately obvious once the file is opened in a hex editor.

5. Entropy and randomness analysis: Encrypted or compressed payloads embedded inside an image increase the local Shannon entropy of the affected pixel regions, since encrypted data statistically resembles random noise far more than natural image data does. Calculating a localized entropy map across the image can highlight suspicious regions of abnormally high randomness that do not correspond to genuinely complex, high-detail areas of the logo (such as sharp edges or fine text).

6. Automated steganalysis tools: Finally, purpose-built steganalysis software such as StegExpose, StegDetect, zsteg, or Stegoveritas automates the above statistical tests, scanning the file against signatures of well-known steganography tools and algorithms and producing a probability score that hidden content is present. Running the suspect logo through several such tools in parallel provides corroborating evidence and reduces the chance of a false negative from relying on any single technique.

Summary: no individual check is proof by itself — a mismatched hash raises suspicion, an abnormal LSB plane or histogram provides statistical evidence, and a positive result from a dedicated steganalysis tool corroborates the finding. Taken together, this layered forensic approach allows investigators to confirm with high confidence whether the company logo has been used to smuggle confidential data.

(b) Two Techniques to Strengthen Security of Hidden Communications

If an organisation legitimately wants to use steganography as one layer of a covert communication channel (for example, to pass sensitive information without drawing attention to the fact that a secret is being transmitted at all), the raw LSB-embedding approach illustrated above is weak on its own. Two practical improvements would substantially raise its resistance to detection and extraction:

1. Encrypt the payload before embedding it (defence in depth): Rather than hiding the secret message in plain, readable form, the sender should first encrypt it with a strong symmetric algorithm such as AES-256 (optionally combined with compression) and only then embed the resulting ciphertext bit-stream into the cover image. This has two compounding benefits: first, ciphertext has near-maximal entropy, similar to natural image noise in the least significant bits, which somewhat masks the statistical anomaly that chi-square tests look for; second, and more importantly, even if a forensic team fully detects and extracts the hidden bit-stream, what they recover is unreadable ciphertext rather than the confidential information itself, so the message content remains protected unless the encryption key is also compromised. This layered 'encrypt-then-hide' approach means an attacker must defeat both the concealment layer and the cryptographic layer to succeed, rather than just one.

2. Use adaptive, key-driven embedding in high-texture regions instead of naive sequential LSB: Simple implementations embed data sequentially into the LSBs of every pixel starting from the top-left corner, which is exactly the predictable pattern that steganalysis tools are tuned to detect. A stronger approach uses an adaptive algorithm (for example, techniques based on the discrete cosine/wavelet transform domain, or edge-adaptive LSB matching) that (i) uses a secret key to pseudo-randomly scatter the embedding positions across the entire image instead of a fixed sequential order, and (ii) preferentially hides data in regions of naturally high texture, noise, or colour variation (such as edges and fine detail in the logo) where small pixel changes are statistically indistinguishable from the image's own natural variation. Together, the randomized position selection and the noise-adaptive placement make the resulting statistical footprint far closer to that of an untouched image, significantly reducing the success rate of chi-square, histogram, and machine-learning-based steganalysis.

Applying both techniques together — encrypting the payload before embedding it with a key-driven, texture-adaptive algorithm — gives the organisation a defence-in-depth covert channel: detection becomes statistically much harder, and even a successful detection and extraction does not, by itself, compromise the confidential information.

2) Legacy Encryption in Corporate Communication

Scenario: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as:
"WECRLTEERDSOEFEAOCAIVDEN".

Transposition ciphers work on a different principle from substitution ciphers: instead of replacing letters with other letters or symbols, they keep every letter of the plaintext unchanged and simply rearrange the order in which those letters appear, according to some systematic rule (a route, a grid, or a key). The phrase 'three levels of arrangement' is the classic description of a Rail Fence Cipher written across three rails (rows), which is one of the simplest and historically best-known route transposition ciphers.

(a) Determining the Most Probable Original Message

In a rail fence cipher, the plaintext is written diagonally in a zig-zag pattern down and up across the chosen number of rails, and the ciphertext is produced by reading the letters off rail by rail (first everything on rail 1, then rail 2, then rail 3, and so on). To decrypt, the process must be reversed: the ciphertext is split back into segments matching the length of each rail, and the original zig-zag grid is reconstructed so the message can be read off diagonally.

Step 1 — Determine the rail lengths for 3 rails: For n rails, the zig-zag repeats with a cycle length of $2 \times (n-1)$. Here $n = 3$, so the cycle length is $2 \times (3-1) = 4$ positions per cycle. Walking through all 25 character positions (numbered 1 to 25) and marking which rail each one falls on shows that rail 1 (the top row, hit once every 4 positions) receives 7 characters, rail 2 (the middle row, hit twice every 4 positions because the zig-zag passes through it going both down and up) receives 12 characters, and rail 3 (the bottom row) receives 6 characters. This gives $7 + 12 + 6 = 25$, which exactly matches the length of the given ciphertext — confirming that a 3-rail arrangement is consistent with the data before any letters are even placed.

Step 2 — Split the 25-character ciphertext into three segments of exactly these lengths, in order: Rail 1 (characters 1–7) = "WECRLTE"; Rail 2 (characters 8–19) = "ERDSOEFEAOC"; Rail 3 (characters 20–25) = "AIVDEN".

Step 3 — Rebuild the zig-zag grid: each rail's letters are placed back into their diagonal positions — rail 1 occupies the 'peak' positions of the zig-zag, rail 3 occupies the 'trough' positions, and rail 2 occupies the diagonal positions in between, moving down then up repeatedly. Doing this for all 25 positions reconstructs the following grid (each column is one character position; dots mark rails the zig-zag does not touch at that position):

W	.	.	.	E	.	.	.	C	.	.	.	R	.	.	.	L	.	.	.	T	.	.	.	E
.	E	.	R	.	D	.	S	.	O	.	E	.	E	.	F	.	E	.	A	.	O	.	C	.
.	.	A	.	.	.	I	.	.	.	V	.	.	.	D	.	.	.	E	.	.	.	N	.	.

Step 4 — Read the plaintext: tracing the zig-zag path column by column (top row, then diagonally down to the middle, down to the bottom, back up through the middle, and so on) reproduces the original message:

WE ARE DISCOVERED FLEE AT ONCE

Verification 1 — trying other rail counts: To confirm that 3 rails (not 2 or 4) is the correct arrangement, the same reconstruction procedure can be attempted with other plausible rail counts. Only the 3-rail arrangement produces grammatically valid, readable English; every other tested count produces a meaningless jumble of letters, which corroborates the scenario's statement that the system specifically uses 'three levels of arrangement':

Rails Tried	Result of Reconstruction	Readable?
2	Produces an unreadable jumble with no valid English word boundaries	No
3	Produces 'WEAREDISCOVEREDFLEEATONCE' — clean English text	Yes
4	Produces another unreadable jumble; rail lengths do not align meaningfully	No

Verification 2 — letter frequency comparison: A second, independent check confirms that this ciphertext was produced by transposition rather than substitution. Because transposition never changes the identity of a letter, only its position, the multiset of letters in the ciphertext must be identical to the multiset of letters in the plaintext. Counting the letters in the ciphertext "WECRLTEERDSOEFEAOC AIVDEN" gives exactly the same frequency table as counting the letters in the recovered plaintext "WEAREDISCOVEREDFLEEATONCE" (both are simply anagrams of one another, 25 letters in total, dominated by 7 occurrences of the letter E, which is also the most common letter in English text):

Letter	E	A	R	D	C	O	W	I	S	V	F	L	T	N
Count	7	2	2	2	2	2	1	1	1	1	1	1	1	1

This exact match — rather than a shifted or altered frequency distribution, as would occur with a Caesar or monoalphabetic substitution cipher — is strong structural evidence that only the positions of the letters were changed, consistent with a transposition cipher and with the plaintext derived above.

(b) Why This Method Is Vulnerable in Modern Communication Systems

The decryption above was carried out entirely by hand, with pencil-and-paper reasoning, in only a few minutes and without knowledge of any secret key beyond the rail count. This ease of manual cryptanalysis exposes several fundamental weaknesses that make single-stage transposition ciphers unsuitable for protecting real corporate communications today:

1. Extremely small key space: The entire 'key' of a rail fence cipher is simply the number of rails used. For a short message like this 25-character example, there are only around 20 meaningfully different rail counts to try (from 2 rails up to one less than the message length), and most of them can be immediately discarded because the resulting grid produces gibberish. An attacker — or even an automated script — can

exhaustively try every possible rail count in a fraction of a second and instantly recognize the one output that forms valid English, exactly as demonstrated in the verification table above.

2. Preserves letter frequency and language statistics: Because transposition rearranges positions without altering the letters themselves, the ciphertext inherits the exact letter-frequency profile of the underlying language, as shown in the frequency table above. Any cryptanalyst trained in classical frequency analysis (originally developed to break substitution ciphers) can instantly recognize the characteristic dominance of E, and can further use digraph/trigraph frequency patterns (such as the common English pairing 'TH', 'ER', 'RE') to guide reconstruction, drastically reducing the search effort.

3. Vulnerable to anagramming and known-plaintext attacks: Since every letter of the plaintext is present in the ciphertext, unchanged, an analyst who suspects the general subject matter of a message (for example, that a corporate memo likely contains words like 'MEETING', 'CONFIDENTIAL', or 'REPORT') can search for anagram subsets of the ciphertext that spell such probable words, then use the surrounding letters to fill in the rest — a known-plaintext or probable-word attack that does not even require correctly guessing the rail count first.

4. No confusion and only weak, single-layer diffusion: Modern cryptographic security rests on two properties defined by Shannon: confusion, which obscures the mathematical relationship between key and ciphertext, and diffusion, which spreads the influence of each plaintext symbol across many ciphertext symbols. A single rail fence pass provides some diffusion (each letter moves to a different position) but zero confusion (the letters are never altered), and only one round of a very simple, well-understood transformation. Modern algorithms such as AES apply many rounds of combined substitution and permutation specifically to build up both properties to a cryptographically strong level — something a single legacy transposition step cannot approach.

5. No protection against tampering or impersonation: The scheme provides confidentiality alone (and, as shown, weak confidentiality at that) with no built-in message authentication code, digital signature, or integrity check. An attacker who intercepts and modifies the ciphertext in transit, or who forges a new message entirely, would not be detected by the receiving system, which is unacceptable for sensitive corporate communications that may carry legal or financial weight.

6. Trivial for modern computing power: Even a manual attacker can break this scheme in minutes; a modern computer can automate the rail-count brute-force search, dictionary/anagram matching, and frequency scoring described above to solve such ciphers in well under a second, for messages of essentially any practical length used in business correspondence.

(c) Proposed Enhancement to Improve Confidentiality

The most effective and practical remedy is to stop relying on transposition as the sole protection mechanism and instead adopt a layered, modern cryptographic design:

Short-term enhancement — combine transposition with substitution (a product cipher): As an immediate, low-cost improvement to the legacy system, the existing rail fence step could be combined with a substitution step (for example, first applying a keyed monoalphabetic or polyalphabetic substitution such as Vigenère to the plaintext, and only then transposing the result — an approach historically used in ciphers like the World War I-era ADFGVX cipher). Layering substitution before transposition means that even if an attacker successfully reverses the transposition step, the recovered text is still meaningless ciphertext rather than plaintext, and the direct letter-frequency correlation that made the original scheme trivial to break is broken, since substitution alters the letters and transposition then scatters the altered letters' positions.

Recommended long-term solution — replace the legacy scheme with a modern authenticated symmetric cipher: The company should migrate corporate communications to a standardized, well-vetted algorithm such as AES-256 operating in an authenticated encryption mode (e.g., AES-GCM), combined with a securely managed key-exchange protocol (such as Diffie–Hellman or a public-key infrastructure) rather than a fixed, guessable rail count. This provides strong confusion and diffusion across many encryption rounds, an effectively unbreakable key space (2^{256} possible keys), and — critically — built-in message integrity and authentication so that any tampering or forgery in transit is automatically detected, none of which the legacy transposition-only method can offer.

In summary, moving from a single, small-key-space transposition scheme to a layered product cipher (short term) and ultimately to a modern authenticated symmetric standard like AES-GCM (long term) would bring the confidentiality, integrity, and overall security of the company's internal communications up to a level appropriate for today's threat landscape.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation